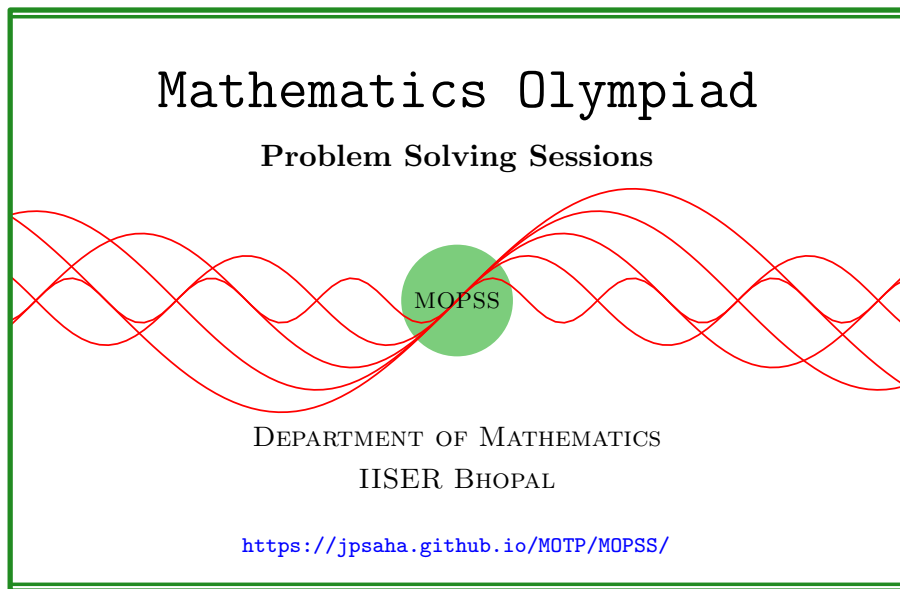


# Functional equations

MOPSS



## Suggested readings

- Evan Chen's advice [On reading solutions](https://blog.evanchen.cc/2017/03/06/on-reading-solutions/), available at <https://blog.evanchen.cc/2017/03/06/on-reading-solutions/>.
- Evan Chen's [Advice for writing proofs/Remarks on English](https://web.evanchen.cc/handouts/english/english.pdf), available at <https://web.evanchen.cc/handouts/english/english.pdf>.
- [Notes on proofs](#) by Evan Chen from [OTIS Excerpts](#) [[Che25](#), Chapter 1].
- [Tips for writing up solutions](https://www.math.utoronto.ca/barbeau/writingup.pdf) by Edward Barbeau, available at <https://www.math.utoronto.ca/barbeau/writingup.pdf>.
- Evan Chen discusses why [math olympiads](#) are a valuable experience for [high schoolers](#) in the post on [Lessons from math olympiads](#), available at <https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/>.

# List of problems and examples

|     |                                                                                                                       |   |
|-----|-----------------------------------------------------------------------------------------------------------------------|---|
| 1.1 | Example (cf. Canadian Mathematical Olympiad 1969 P8, Australian Mathematics Competition 1984, India Pre-RMO 2012 P16) | 2 |
| 1.2 | Exercise (Putnam 1963 A2, AoPS)                                                                                       | 3 |
| 1.3 | Example (India RMO 2006 P7)                                                                                           | 3 |

## §1 Functional equations

**Example 1.1** (cf. Canadian Mathematical Olympiad 1969 P8, Australian Mathematics Competition 1984, India Pre-RMO 2012 P16). [Tao06, Problem 3.2] Suppose

$$x_1, x_2, x_3, x_4, \dots$$

is a sequence of integers satisfying the following properties:

- (1)  $x_2 = 2$ ,
- (2)  $x_{mn} = x_m x_n$  for all positive integers  $m, n$ ,
- (3)  $x_m < x_n$  for any positive integers  $m, n$  with  $m < n$ .

Find  $x_{2024}$ . What is the value of the sum

$$x_1 + x_2 + \dots + x_{2024}?$$

**Summary** — Observe that  $x_{2^n} = 2^n$  for any  $n \geq 1$ . Combining this with the hypothesis that  $\{x_n\}_{n \geq 1}$  is an increasing sequence of **positive integers**, conclude that  $x_n = n$  for any  $n \geq 1$ .

**Walkthrough** —

- (a) What can be said about  $x_4, x_8, x_{16}, x_{32}$ ?
- (b) Note that  $x_4 = x_{2 \times 2}, x_8 = x_{4 \times 2}, x_{16} = x_{8 \times 2}, x_{32} = x_{16 \times 2}$ .
- (c) Can one show that  $x_{2^n} = 2^n$  for any  $n \geq 1$ ?
- (d) Show that  $x_m = m$  for any  $m \geq 1$  (does property (3) help?).

**Solution 1.** From the second condition, we obtain

$$x_{2^n} = x_2^n$$

for any integer  $n \geq 1$ . Using the first condition, it follows by induction that

$$x_{2^n} = 2^n$$

for any integer  $n \geq 1$ . Since  $\{x_n\}_{n \geq 1}$  is an increasing sequence of positive integers, it follows that  $x_n = n$  for any positive integer  $n$ . This gives

$$x_{2024} = 2024.$$

It follows that

$$x_1 + x_2 + \cdots + x_{2024} = 1012 \cdot 2025 = 2049300.$$

■

**Exercise 1.2** (Putnam 1963 A2, AoPS). Let  $f: \mathbb{N} \rightarrow \mathbb{N}$  be a strictly increasing function such that  $f(2) = 2$  and  $f(mn) = f(m)f(n)$  for every relatively prime pair of positive integers  $m$  and  $n$ . Show that  $f(n) = n$  for every positive integer  $n$ .

**Walkthrough** —

(a)

**Solution 2.**

■

**Example 1.3** (India RMO 2006 P7). Let  $X$  be the set of all positive integers greater than or equal to 8, and let  $f: X \rightarrow X$  be a function such that  $f(x+y) = f(xy)$  for all  $x \geq 4, y \geq 4$ . If  $f(8) = 9$ , determine  $f(9)$ .

**Walkthrough** — Using the given condition, try to express  $f(9)$  in terms of  $f(8)$ .

**Solution 3.** Note that

$$f(9) = f(4+5) = f(20) = f(4+16) = f(64),$$

and

$$f(8) = f(4+4) = f(16) = f(8+8) = f(64)$$

hold. Using the given condition  $f(8) = 9$ , it follows that  $f(9) = 9$ .

■

## References

- [Che25] EVAN CHEN. *The OTIS Excerpts*. Available at <https://web.evanchen.cc/excerpts.html>. 2025, pp. vi+289 (cited p. 1)
- [Tao06] TERENCE TAO. *Solving mathematical problems*. A personal perspective. Oxford University Press, Oxford, 2006, pp. xii+103. ISBN: 978-0-19-920560-8; 0-19-920560-4 (cited p. 2)